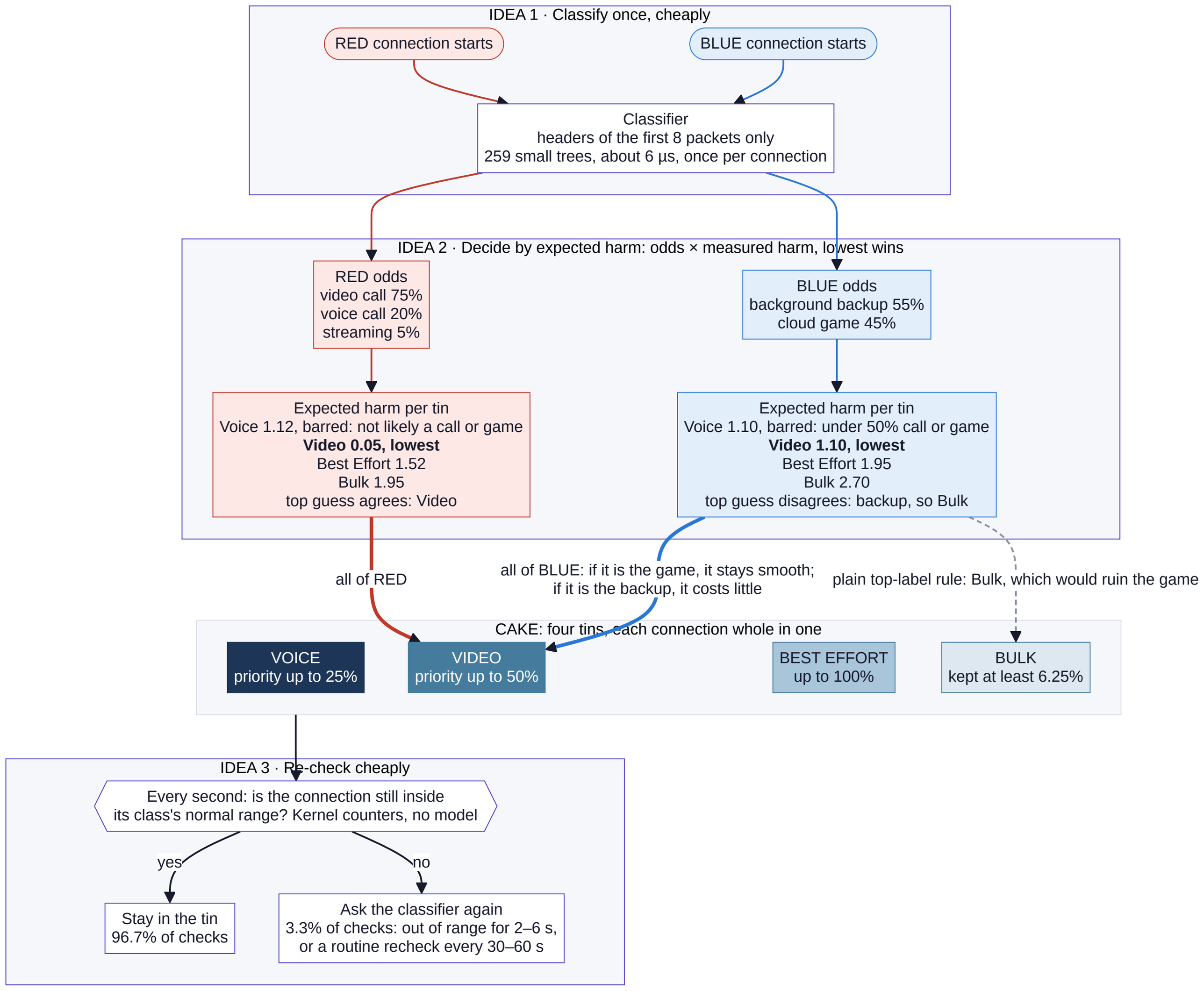
IDEA 1 · Classify once, cheaply
RED connection starts
BLUE connection starts
Classifier
headers of the first 8 packets only
259 small trees, about 6 µs, once per connection
IDEA 2 · Decide by expected harm: odds × measured harm, lowest wins
RED odds
video call 75%
voice call 20%
streaming 5%
BLUE odds
background backup 55%
cloud game 45%
Expected harm per tin
Voice 1.12, barred: not likely a call or game
Video 0.05, lowest
Best Effort 1.52
Bulk 1.95
top guess agrees: Video
Expected harm per tin
Voice 1.10, barred: under 50% call or game
Video 1.10, lowest
Best Effort 1.95
Bulk 2.70
top guess disagrees: backup, so Bulk
all of RED
all of BLUE: if it is the game, it stays smooth;
if it is the backup, it costs little
plain top-label rule: Bulk, which would ruin the game
CAKE: four tins, each connection whole in one
VOICE
priority up to 25%
VIDEO
priority up to 50%
BEST EFFORT
up to 100%
BULK
kept at least 6.25%
IDEA 3 · Re-check cheaply
Every second: is the connection still inside
its class's normal range? Kernel counters, no model
yes
no
Stay in the tin
96.7% of checks
Ask the classifier again
3.3% of checks: out of range for 2–6 s,
or a routine recheck every 30–60 s